

|                  |                                                                    |
|------------------|--------------------------------------------------------------------|
| <b>Exp No: 1</b> | <b>Study of various Network commands used in Linux and Windows</b> |
|------------------|--------------------------------------------------------------------|

**AIM:**

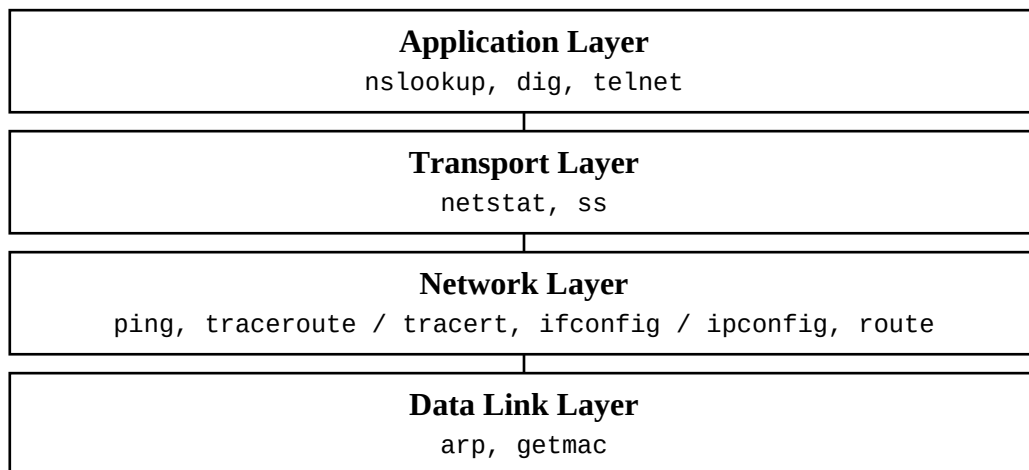
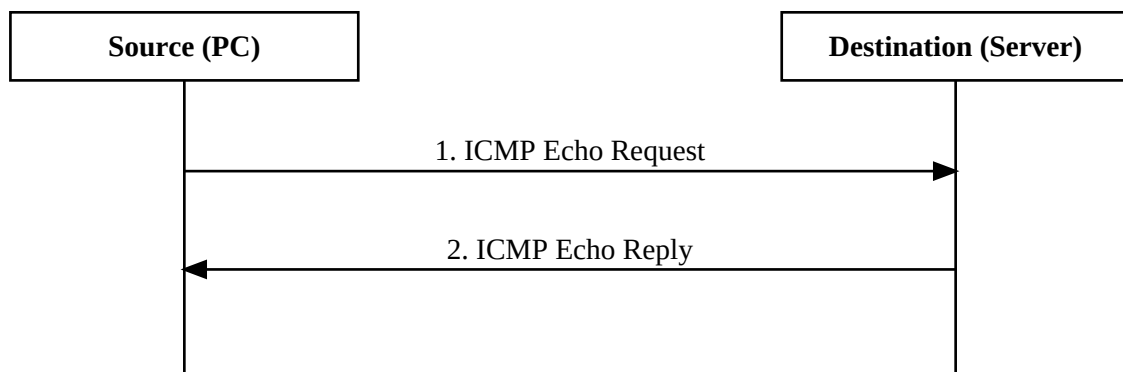
To study and practise the various network commands used in Linux and Windows operating systems.

**ALGORITHM / PROCEDURE:**

1. Open the Terminal in Linux and the Command Prompt (cmd) in Windows.
2. Use hostname to see the name of the computer.
3. Use ifconfig (Linux) and ipconfig (Windows) to see the IP address, subnet mask and MAC address.
4. Use ping to check whether the remote system is reachable. Note the packet loss and the time taken.
5. Use traceroute (Linux) and tracert (Windows) to see the path taken by the packets.
6. Use nslookup and dig to find the IP address of a domain name.
7. Use netstat to see the active connections and open ports.
8. Use arp -a to see the IP address to MAC address table.
9. Use route (Linux) and route print (Windows) to see the routing table.
10. Note down the output of each command.

**Commands used:**

| <b>Purpose</b>     | <b>Linux</b>          | <b>Windows</b>     |
|--------------------|-----------------------|--------------------|
| Host name          | hostname              | hostname           |
| IP configuration   | ifconfig / ip addr    | ipconfig /all      |
| Check connectivity | ping -c 4 google.com  | ping google.com    |
| Trace the route    | traceroute google.com | tracert google.com |
| DNS lookup         | nslookup / dig        | nslookup           |
| Active connections | netstat -tuln         | netstat -an        |
| ARP table          | arp -a                | arp -a             |
| Routing table      | route -n              | route print        |
| MAC address        | ip link               | getmac             |

**DIAGRAM:****Fig 1: Network commands and the layers they work in****Fig 2: Working of ping command****OUTPUT:****Linux:**

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ hostname
USER -VirtualBox
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.1.10 netmask 255.255.255.0 broadcast 192.168.1.255
    inet6 fe80::a00:27ff:fe4e:66a1 prefixlen 64 scopeid 0x20<link>
    ether 08:00:27:4e:66:a1 txqueuelen 1000 (Ethernet)
    RX packets 2543 bytes 2154321 (2.1 MB)
    TX packets 1987 bytes 312456 (312.4 KB)

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ ping -c 4 google.com
PING google.com (142.250.195.78) 56(84) bytes of data.
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=1 ttl=117 time=14.2 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=2 ttl=117 time=14.8 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=3 ttl=117 time=13.9 ms
64 bytes from maa03s45-in-f14.1e100.net (142.250.195.78): icmp_seq=4 ttl=117 time=15.2 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3005ms
rtt min/avg/max/mdev = 13.904/14.525/15.230/0.512 ms
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ traceroute google.com
traceroute to google.com (142.250.195.78), 30 hops max, 60 byte packets
 1 _gateway (192.168.1.1) 1.102 ms 1.045 ms 0.987 ms
 2 10.20.0.1 (10.20.0.1) 4.321 ms 4.287 ms 4.410 ms
 3 172.16.5.9 (172.16.5.9) 8.765 ms 8.801 ms 8.730 ms
 4 72.14.215.85 (72.14.215.85) 12.543 ms 12.498 ms 12.610 ms
 5 maa03s45-in-f14.1e100.net (142.250.195.78) 14.230 ms 14.187 ms 14.301 ms
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ nslookup google.com
Server:      192.168.1.1
Address:     192.168.1.1#53

Non-authoritative answer:
Name:   google.com
Address: 142.250.195.78
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ dig google.com +short
142.250.195.78
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ netstat -tuln
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22               0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631            0.0.0.0:*               LISTEN
tcp6       0      0 :::80                   :::*                    LISTEN
udp        0      0 0.0.0.0:68              0.0.0.0:*
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ arp -a
_gateway (192.168.1.1) at c4:e9:84:2a:5b:10 [ether] on enp0s3
? (192.168.1.12) at 3c:52:82:7d:11:9e [ether] on enp0s3
```

```
USER @USER -VirtualBox: ~
```

```
USER @USER -VirtualBox:~$ route -n
Kernel IP routing table
Destination Gateway Genmask Flags Metric Ref Use Iface
0.0.0.0 192.168.1.1 0.0.0.0 UG 100 0 0 enp0s3
192.168.1.0 0.0.0.0 255.255.255.0 U 100 0 0 enp0s3
```

**Windows:**

## Command Prompt

```
C:\Users\USER >hostname
DESKTOP-USER
```

## Command Prompt

```
C:\Users\USER >ipconfig /all
```

## Windows IP Configuration

```
Host Name . . . . . : DESKTOP-USER
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
```

## Ethernet adapter Ethernet:

```
Physical Address. . . . . : 00-1A-2B-3C-4D-5E
DHCP Enabled. . . . . : Yes
IPv4 Address. . . . . : 192.168.1.25(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Default Gateway . . . . . : 192.168.1.1
DHCP Server . . . . . : 192.168.1.1
DNS Servers . . . . . : 8.8.8.8
```

## Command Prompt

```
C:\Users\USER >ping google.com
```

```
Pinging google.com [142.250.195.78] with 32 bytes of data:
Reply from 142.250.195.78: bytes=32 time=15ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=14ms TTL=117
Reply from 142.250.195.78: bytes=32 time=16ms TTL=117
```

## Ping statistics for 142.250.195.78:

```
Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
Minimum = 14ms, Maximum = 16ms, Average = 14ms
```

## Command Prompt

```
C:\Users\USER >tracert google.com
```

```
Tracing route to google.com [142.250.195.78]
over a maximum of 30 hops:
```

```
 1  <1 ms  <1 ms  <1 ms  192.168.1.1
 2    4 ms   4 ms   4 ms  10.20.0.1
 3    9 ms   8 ms   9 ms  172.16.5.9
 4   13 ms  12 ms  12 ms  72.14.215.85
 5   14 ms  14 ms  15 ms  142.250.195.78
```

```
Trace complete.
```

## Command Prompt

```
C:\Users\USER >nslookup google.com
Server:  dns.google
Address:  8.8.8.8
```

## Non-authoritative answer:

```
Name:     google.com
Address:  142.250.195.78
```

## Command Prompt

```
C:\Users\USER >netstat -an
```

## Active Connections

| Proto | Local Address      | Foreign Address    | State       |
|-------|--------------------|--------------------|-------------|
| TCP   | 0.0.0.0:135        | 0.0.0.0:0          | LISTENING   |
| TCP   | 0.0.0.0:445        | 0.0.0.0:0          | LISTENING   |
| TCP   | 192.168.1.25:50234 | 142.250.195.78:443 | ESTABLISHED |
| UDP   | 0.0.0.0:5353       | *.*                |             |

## Command Prompt

```
C:\Users\USER >arp -a
```

```
Interface: 192.168.1.25 --- 0x8
```

| Internet Address | Physical Address  | Type    |
|------------------|-------------------|---------|
| 192.168.1.1      | c4-e9-84-2a-5b-10 | dynamic |
| 192.168.1.255    | ff-ff-ff-ff-ff-ff | static  |

## Command Prompt

```
C:\Users\USER >route print
```

## IPv4 Route Table

```
=====
```

Active Routes:

| Network | Destination | Netmask       | Gateway     | Interface    | Metric |
|---------|-------------|---------------|-------------|--------------|--------|
|         | 0.0.0.0     | 0.0.0.0       | 192.168.1.1 | 192.168.1.25 | 25     |
|         | 127.0.0.0   | 255.0.0.0     | On-link     | 127.0.0.1    | 331    |
|         | 192.168.1.0 | 255.255.255.0 | On-link     | 192.168.1.25 | 281    |

## Command Prompt

```
C:\Users\USER >getmac
```

| Physical Address  | Transport Name                                    |
|-------------------|---------------------------------------------------|
| 00-1A-2B-3C-4D-5E | \Device\NPF{8F3A21B0-5C6D-4E7F-9A1B-2C3D4E5F6A7B} |

**RESULT:**

Thus the various network commands used in Linux and Windows were studied and executed successfully.